

# Soumik Pramanik

Tokyo, Japan | soumikpramanik28@gmail.com | +8107085810425, +917003486044 | LinkedIn

## Summary

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Experienced Data scientist and AI Engineer with a strong background in building scalable data pipelines, developing machine learning models and delivering impactful software solutions. Skilled in Python, SQL, PostgreSQL, and end-to-end project deployment. The first author of International Springer Nature publication (**Springer Nature Switzerland**) on a new image dataset along with classification and biasing techniques, demonstrating strong analytical and research skills.

## Experience

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### Autocomplete Search suggestion for Rakuten Music.

Jan 2024

**Data Scientist**, Rakuten – Tokyo, JP

- Design a query summary framework to collect and analyze key statistics per query, enabling the generation of candidate suggestions for an Autocomplete service.
- Designed and implemented data collection pipelines (Python, Hadoop, PySpark) to extract essential features for training machine learning models, optimizing the ranking and relevance of Autocomplete suggestions.
- The ranking algorithm has been **improved by 17%**, resulting in a **39% enhancement in overall performance** compared to the previous results.

**Technologies:** Python, SQL, PySpark, Scikit-learn, TensorFlow

### GSM Analyzer.

- Architected and deployed a production-grade analytics platform to monitor, diagnose, and prioritize GSM call failures, driving data-driven improvements in network reliability and operational efficiency, supported by interactive BI dashboards for rapid analysis and decision-making.
- Implemented a scoring-based ranking system to prioritize failure codes by traffic impact and KPI degradation, enabling operations teams to focus on the most business-critical issues first.
- Enabled faster root-cause analysis and decision-making, reducing time-to-resolution for high-impact network incidents and supporting continuous improvement of network performance.

**Technologies:** BigQuery, Power Automate, Airflow, Python, DOMO

### Autonomous Multi-Agent AI Platform.

- Architected and led the design and development of an autonomous multi-agent AI platform, defining agent orchestration, governance, and scalable deployment patterns for AI-driven analytics and decision automation.
- Designed end-to-end AI workflows integrating RAG, dynamic SQL generation, agentic data science tools and automated insight generation across heterogeneous data sources, ensuring traceability, reliability, and production readiness.
- Established MLOps and lifecycle standards (API services, CI/CD, monitoring, failure handling) to deliver secure, scalable, and continuously operated AI services.

**Technologies:** Python, SQL, FastAPI, MCP, Docker, Airflow, GCP BigQuery, LlamaIndex, Langgraph, Agno

## Publications

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### Fish92: A Novel Dataset for Indigenous Fish Classification

Dec 2024

**Soumik Pramanik**, Shrihari A, Prithwjit Guha

9th International conference on Computer vision & Image processing (CVIP), IIITDM Kancheepuram, India, 2024

**Cham: Springer Nature Switzerland, 2025**

[doi.org/10.1007/978-3-031-93694-4\\_19](https://doi.org/10.1007/978-3-031-93694-4_19)

**Highlights:** Fish92 Dataset, Feature Extraction, Classification Models, Debiasing Techniques

## Education

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**IIT Guwahati** - MTech in Robotics and Artificial Intelligence

Sept 2021 – May 2023

- GPA: 8.60/10.0

• **Coursework:** Image processing, Machine Learning, Database Management Systems

**Govt. College of Engineering and Leather Technology** - BTech in Computer

Aug 2015 – May 2019

Science

- GPA: 8.45/10.0

• **Coursework:** Operating Systems, Computer Networks

## Skills/Technologies

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**Languages:** C++, C, Python, SQL

**Frameworks/Tools:** Langgraph, Agno, Pyspark, Scikit-learn, TensorFlow, PyTorch

**MLOps/Tools:** FastAPI, Docker, Airflow, Git, GCP

**Database:** postgresql

## Achievements

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- Awarded **Best Performer** in RM QCC 2025 (1H) for the project: *IMS Call Setup RCA*.
- Achieved **Top 1% (99th percentile)** in GATE 2021; qualified GATE consecutively in 2019, 2020, and 2021.